QUASI-MODULARITY OF THE SL3 LIE-THEORETIC Q-MZV FAMILY VIA CONSTANT TERMS

ANTUN MILAS

Allowed standard prerequisites (editorial).

- Standard q-binomial summation, vertex-algebra character definitions and classical elliptic/theta functions. Original source bibliography, including Lang, Arakawa and AM where relevant.

Source reference. Original reference 3.14: algebra-geometry/sources/sigma-2026-087/native/sigma26-087.tex, source lines 1402–1402. Cross-reference gives an introductory specialization, earlier narrative comparison or section pointer; all source-local arguments used for the counted theorem are included.

Denote by Δ a root system of ADE type, by Δ_+ the set of positive roots, and let $\langle\cdot,\cdot\rangle$ denote the inner product normalized such that $\langle\alpha,\alpha\rangle = 2$ for every root α. For integers $k_\alpha \geq 1$, we define a q-MZV associated with $\mathfrak{g}$ as

$$\zeta_{\mathfrak{g},q}(k_\alpha;\alpha\in\Delta_+) := \sum_{\lambda\in P_+} \frac{q^{\frac{1}{2}\sum_{\alpha\in\Delta_+} k_\alpha\langle\lambda+\rho,\alpha\rangle}}{\prod_{\alpha\in\Delta_+}(1-q^{\langle\lambda,\alpha+\rho\rangle})^{k_\alpha}}, \tag{0.1}$$

where the summation is over the cone of positive dominant integral weights. The asymptotic behavior motivates our definition. Indeed, after setting $q = \mathrm{e}^{2\pi\mathrm{i}\tau}$ and taking the asymptotic limit $\tau \to 0+$ with $k_\alpha := k \geq 2$, we recover Witten's ζ-function of $\mathfrak{g}$ (see [51])

$$\sum_{\lambda\in P_+} \frac{1}{\dim(V_\lambda)^k},$$

as the leading coefficient of the asymptotic expansion. This fact follows from the Weyl dimension formula.

In the simplest case of $\mathfrak{g} = \mathfrak{sl}_2$, where $\Delta = \{\pm\alpha\}$, these q-zeta values are given by

$$\zeta_{\mathfrak{sl}_2,q}(k) = \sum_{n\geq 1} \frac{q^{\frac{k}{2}n}}{(1-q^n)^k}.$$

This slightly non-standard q-zeta value has several nice properties. For instance, the summand is invariant under $q \to q^{-1}$ for k even and skew-invariant for k odd. For simplicity, we only consider k even here. Then it is known (or it can be easily verified) that for $k \geq 1$, $\zeta_{\mathfrak{sl}_2,q}(2k)$, up to an additive constant, is a linear combination of Eisenstein series, and in fact we have

$$\mathbb{C}[\zeta_{\mathfrak{sl}_2,q}(2k) : k \geq 1] = \underbrace{\mathbb{C}[E_2(q), E_4(q), E_6(q)]}_{:=\mathcal{QM}},$$

where E_{2i} are Eisenstein series defined with their q-expansion; see formula (1.1) in Section 1. The space $\mathcal{QM}$ is called the ring of (holomorphic) quasi-modular forms on $\mathrm{SL}(2,\mathbb{Z})$.

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1. Multiple q-zeta values, quasi-modular forms, and characters

This section contains the second part of the paper. Throughout, we let $\tau \in \mathbb{H}$, $z \in \mathbb{C}$, $q = \mathrm{e}^{2\pi \mathrm{i}\tau}$, and $\zeta = \mathrm{e}^{2\pi \mathrm{i}z}$. Unless stated otherwise, all identities involving Fourier series in ζ are interpreted in the annulus specified in the statement, which determines the corresponding geometric-series expansion of each denominator. We begin by recalling the elliptic and quasi-Jacobi functions needed for our constant-term computations and then apply these techniques to characters of genus-zero class-$\mathcal{S}$ vertex operator algebras.

1.1. Elliptic functions, theta functions, and quasi-modular forms.

We present some basic (and mostly well-known) results on elliptic functions, (quasi)-Jacobi forms, and their Fourier coefficients. We closely follow [34].

The Jacobi theta function is given by

$$\vartheta(z;\tau) = \sum_{n\in\mathbb{Z}+\frac{1}{2}} \mathrm{e}^{\pi \mathrm{i} n^2\tau + 2\pi \mathrm{i} n(z+\frac{1}{2})}.$$

We also define Eisenstein series with their q-expansion, for integers $k \geq 1$ (and with the usual convention that E_2 is quasimodular)

$$G_{2k}(\tau) = 2\zeta(2k) + \frac{(2\pi \mathrm{i})^{2k}}{(2k-1)!}\sum_{n\geq 1}\frac{n^{2k-1}q^n}{1-q^n}, \qquad q = \mathrm{e}^{2\pi \mathrm{i}\tau},$$

and slightly normalized versions

$$E_{2k}(\tau) = 1 - \frac{4k}{B_{2k}}\sum_{n\geq 1}\frac{n^{2k-1}q^n}{1-q^n}, \tag{1.1}$$

where B_{2k} is the Bernoulli number. Recall that $\mathcal{QM}$ us the ring of quasi-modular forms for the full modular group discussed earlier. A famous result of S. Ramanujan is that

$$\left(q\frac{\mathrm{d}}{\mathrm{d}q}\right):\ \mathcal{QM} \to \mathcal{QM},$$

so $\mathcal{QM}$ is a differential ring. This well-known fact is employed throughout this section and in later parts.

The Weierstrass ζ-function[1] and the $\wp$-function are defined as

$$\zeta(z;\tau) = \frac{1}{z} + \sum_{(n,m)\neq(0,0)}\left(\frac{1}{z-m\tau-n} + \frac{1}{m\tau+n} + \frac{z}{(m\tau+n)^2}\right),$$

$$\wp(z;\tau) = \frac{1}{z^2} + \sum_{(n,m)\neq(0,0)}\frac{1}{(z-m-n\tau)^2} - \frac{1}{(m+n\tau)^2},$$

$$\wp'(z;\tau) = -\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}z}\wp(z;\tau).$$

Their Laurent expansions around $z = 0$ are then given by

$$\zeta(z;\tau) = \frac{1}{z} - \sum_{n\geq 1} G_{2n+2}(\tau) z^{2n+1},$$

$$\wp(z;\tau) = \frac{1}{z^2} + \sum_{n\geq 1}(2n+1)G_{2n+2}(\tau)z^{2n},$$

$$\wp'(z;\tau) = \frac{1}{z^3} - \sum_{n\geq 1} n(2n+1)G_{2n+2}(\tau)z^{2n-1}.$$

[1]Not to be confused with the Fourier coefficient $\zeta = \mathrm{e}^{2\pi \mathrm{i}z}$ used below.

For concrete computation, it is convenient to use the so-called P-functions or "propagators" (widely used for computations of genus one correlators in CFT and vertex operator algebra [52]):

$$P_k(\zeta;q)=\frac{(2\pi\mathrm{i})^k}{(k-1)!}\sum_{n\geq 1}\left(\frac{n^{k-1}\zeta^n q^n}{1-q^n}+(-1)^k\frac{n^{k-1}\zeta^{-n}}{1-q^n}\right).$$

Then inside $|q|<\left|\zeta^{-1}\right|<1$ we have

$$P_1\big(\mathrm{e}^{2\pi\mathrm{i}z},\tau\big)=-\zeta(z;\tau)+G_2(\tau)z-\pi\mathrm{i}, \tag{1.2}$$

$$P_2\big(\mathrm{e}^{2\pi\mathrm{i}z},\tau\big)=\wp(z,\tau)+G_2(\tau), \tag{1.3}$$

$$P_3\big(\mathrm{e}^{2\pi\mathrm{i}z},\tau\big)=-\wp'(z,\tau). \tag{1.4}$$

Similarly, $P_k\big(\mathrm{e}^{2\pi\mathrm{i}z};\tau\big)$, $k\geq 4$, is proportional to the $(k-2)$-th z-derivative of $\wp(z;\tau)$. The second and the third relations (1.3)–(1.4) are equivalent to the following:

$$\wp(z,\tau)=-G_2(\tau)+(2\pi\mathrm{i})^2\sum_{n\geq 1}\left(\frac{nq^n\zeta^n}{(1-q^n)}+\frac{n\zeta^{-n}}{(1-q^n)}\right),$$

$$\wp'(z,\tau)=-(2\pi\mathrm{i})^3\sum_{n\geq 1}\left(\frac{n^2q^n\zeta^n}{(1-q^n)}-\frac{n^2\zeta^{-n}}{(1-q^n)}\right).$$

To see that (1.2) holds, we require an identity

$$-\frac{1}{2}+\frac{\zeta}{\zeta-1}-\sum_{n\geq 1}\left(\frac{\zeta^{-1}q^n}{1-\zeta^{-1}q^n}-\frac{q^n\zeta}{1-q^n\zeta}\right)=\frac{1}{2}+\sum_{n\geq 1}\left(\frac{\zeta^nq^n}{1-q^n}-\frac{\zeta^{-n}}{1-q^n}\right), \tag{1.5}$$

again valid in $|q|<|\zeta^{-1}|<1$. We denote either side of (1.5) by $J_1(z;\tau)$. This function does not transform as a Jacobi form, instead we have

$$J_1(z+m\tau+n;\tau)=J_1(z;\tau)-m,$$

$$J_1\left(\frac{z}{\tau};\frac{-1}{\tau}\right)=\tau J_1(z;\tau)+z.$$

We also require $\sigma(z;\tau)$, which can be defined using the theta function [34]

$$\sigma(z;\tau)=-\exp\left(\frac{\eta_1 z^2}{2}\right)\frac{\vartheta(z;\tau)}{2\pi\eta(\tau)^3},$$

where $\eta_1=G_2(\tau)$ in the normalization of [34]; for our purposes, it is a function of τ only and will not be used elsewhere.

Functions $\wp$, ϑ, and σ are related through a fundamental relation [34]:

$$\wp(z_1;\tau)-\wp(z_2;\tau)=\frac{\sigma(z_1+z_2;\tau)\sigma(z_1-z_2;\tau)}{\sigma(z_1;\tau)^2\sigma(z_2;\tau)^2}=\frac{\vartheta(z_1+z_2;\tau)\vartheta(z_1-z_2;\tau)}{4\pi^2\vartheta(z_1;\tau)^2\vartheta(z_2;\tau)^2\eta(\tau)^6}. \tag{1.6}$$

Our next result concerns the constant term in the Fourier expansion of elliptic functions with respect to $\zeta=\mathrm{e}^{2\pi\mathrm{i}z}$. This result is well known and has appeared previously in the literature; see, for example, [26, 40]. In the present work, however, we adopt the approach of [14], where all Fourier coefficients of certain index-zero Jacobi forms are investigated, rather than focusing solely on the constant term.

Proposition 1.1. *Let $f(z;\tau)\in\mathbb{C}[\wp(z;\tau),\wp'(z;\tau)]$. Then, in the range $|q|<\left|\zeta^{-1}\right|<1$, the constant term $\mathrm{CT}_\zeta f(z;\tau)$ belongs to $\mathcal{QM}$, and the non-constant terms $\mathrm{CT}_\zeta f(z;\tau)\zeta^{-r}$, $r\neq 0$, are not quasi-modular.*

Proof. We may assume that $f = \wp^r(\wp')^s$. Using the Weierstrass equation

$$(\wp')^2 = 4\wp^3 - g_2(\tau)\wp - g_3(\tau),$$

where $g_2(\tau)$ and $g_3(\tau)$ are multiples of $E_4(\tau)$ and $E_6(\tau)$, respectively, every higher power of $\wp'$ can be reduced to the cases $s = 0$ and $s = 1$. We use the Laurent expansion in the variable z around zero to write

$$f(z;\tau) = \sum_{0\leq i<\frac{\ell}{2}} \frac{D_{\ell-2i}(\tau)}{z^{\ell-2i}} + D_0(z) + O(z),$$

where the z-coefficients $D_{\ell-2i}(\tau)$ are holomorphic modular forms of weight $2i$ that can be explicitly computed using the Laurent expansion of $\wp$ and $\wp'$, and $\ell = 2r+3s$. Of course, $D_0(z)$ could be zero depending on the parity of s. To compute the ζ-constant term, we proceed as in [14, Theorem 1.1] dealing with a particular Jacobi form of index zero (the case when m is even in [14] is precisely the case of elliptic functions here).

For the nonzero Fourier coefficient $f_r(\tau)$, $r \neq 0$, we compute the contour integral

$$\int_{\partial P_{z_0}} f(w;\tau)\mathrm{e}^{-2\pi \mathrm{i} r w}\,\mathrm{d}w,$$

where

$$P_{z_0} = \{z_0 + r\tau + s \mid 0 \leq r, s \leq 1\},$$

in two different ways: directly as a contour integral, and by the residue theorem by choosing

$$z_0 = -\tfrac{1}{2} - \tfrac{\tau}{2},$$

so that the only pole inside the contour is at $z = 0$. This quickly leads to a formula for the r-th coefficient, which is quasi-modular if and only if $r = 0$ and not quasi-modular otherwise (see [14, Theorem 1.1]). More explicitly,

$$\mathrm{CT}_\zeta f(z;\tau) = D_0(\tau) + \sum_i \lambda_i D_{2i}(\tau)E_{2i}(\tau),$$

where λ_i are some coefficients. This is certainly contained in $\mathcal{QM}$. □

1.2. **Quasi-Jacobi forms.** We are also interested in Fourier coefficients, or more precisely the constant term, of more general functions that include τ-derivatives of elliptic functions. For that we need the concept of *quasi-Jacobi forms* (see [37, 42, 50] for instance). We are only concerned with the subspace of meromorphic quasi-Jacobi forms of index 0 with poles at $L_\tau = \mathbb{Z}\tau + \mathbb{Z}$. These are known to form a subring $\mathcal{QJ}^0 := \mathbb{C}[E_6, E_4, J_1, \wp, \wp']$ inside the ring of quasi-Jacobi forms $\mathcal{QJ} := \mathbb{C}[F, E_6, E_4, J_1, \wp, \wp']$ (E_6 can be omitted from both rings though), where F is a particular quasi-Jacobi form of non-zero index not needed here [42]. Besides $J_1(z;\tau)$ mentioned earlier, we also require more general quasi-Jacobi forms ($k \geq 2$):

$$J_k(z;\tau) = B_k - k\sum_{n,r\geq 1} r^{k-1}\big(\zeta^n + (-1)^k\zeta^{-n}\big)q^{nr}.$$

Next we define the polynomials $p_m(x)$, $m \geq 2$ via

$$\sum_{n\geq 1} n^{m-1}x^n = \frac{xp_m(x)}{(1-x)^m}.$$

For instance, $p_2(x) = 1$, $p_3(x) = 1 + x$, etc. These are known as (slightly shifted) Euler's polynomials. The next result is elementary and essentially follows from the definition of $p_m(x)$.

Lemma 1.2. *We have*

(1) *For every even $s \geq 0$, and $k \geq 1$,*

$$\sum_{r\geq 1} \frac{r^s q^r p_{2k}(q^r)}{(1-q^r)^{2k}} \in \mathcal{QM}.$$

(2) *For every odd $s \geq 1$, and $k \geq 1$,*

$$\sum_{r\geq 1} \frac{r^s q^r (1+q^r) p_{2k}(q^r)}{(1-q^r)^{2k+1}} \in \mathcal{QM}.$$

Proof. For (1), it is enough to observe:

$$\sum_{r\geq 1} r^j \frac{q^r p_{a+j}(q^r)}{(1-q^r)^{a+j}} = \sum_{r\geq 1,\, n\geq 1} r^j n^{a+j-1} q^{nr} = \left(q\frac{\mathrm{d}}{\mathrm{d}q}\right)^j \sum_{r\geq 1,\, n\geq 1} n^{a-1} q^{nr} = \left(q\frac{\mathrm{d}}{\mathrm{d}q}\right)^j \sum_{n\geq 1} \frac{n^{a-1}q^n}{1-q^n}.$$

This implies that, for every s and k as above, we may choose $a \geq 2$ and $j \geq 0$ such that the sum in (1) is a (higher) derivative of an Eisenstein series, or an Eisenstein series with its constant term removed. Similarly, any series appearing in (2) can be expressed as a linear combination of derivatives of two Eisenstein series. In either case, the resulting sum lies in $\mathcal{QM}$. □

Theorem 1.3. *Let $f(z;\tau) \in \mathcal{QJ}^0$. Then in the range $|q| < \left|\zeta^{-1}\right| < 1$, $\mathrm{CT}_\zeta f(z;\tau) \in \mathcal{QM}$.*

Proof. Compared with Proposition 1.1, where f is doubly periodic in z and belongs to $\mathbb{C}[\wp, \wp']$, the only new issue is the contribution of $J_1(z;\tau)$ and thus it suffices to prove the statement for

$$f(z;\tau) = J_1(z;\tau)^j g(z;\tau),$$

where $g(z;\tau) \in \mathbb{C}[\wp, \wp']$ is elliptic and $j \geq 1$. We use induction on j. For $j = 1$, we have to analyze

$$J_1(z;\tau) g(z;\tau),$$

where $g(z;\tau)$ is elliptic. We will compute the constant term by expanding $g(z;\tau) = \zeta^r g_r(\tau)$, where $g_0(\tau)$ is quasi-modular (possibly zero) and non-zero Fourier coefficients are given by [14]

$$g_r(\tau) = \frac{q^r}{1-q^r} \sum_{0\leq \ell < m/2} r^{m-2\ell-1} D_\ell(\tau), \qquad r > 0,$$

and

$$g_r(\tau) = -\frac{1}{1-q^{-r}} \sum_{0\leq \ell < m/2} r^{m-2\ell-1} D_\ell(\tau), \qquad r < 0,$$

where the parity of m depends on the parity of the z powers in the Laurent expansion of $g(z;\tau)$, and $D_\ell(\tau)$ are modular forms as in Proposition 1.1. Multiplying out $J_1(z;\tau) = \frac{1}{2} + \sum_{n\geq 1}\left(\frac{\zeta^n q^n}{1-q^n} - \frac{\zeta^{-n}}{1-q^n}\right)$ and $g(z;\tau)$, we quickly find the constant term of $J_1(z;\tau)g(z;\tau)$ to be

$$\frac{1}{2} g_0(\tau) - \sum_{0\leq \ell < m/2} D_\ell(\tau) \sum_{r\geq 1} \frac{r^{m-2\ell-1} q^r}{(1-q^r)^2} - \sum_{0\leq \ell < m/2} D_\ell(\tau) \sum_{r\geq 1} \frac{(-r)^{m-2\ell-1} q^r}{(1-q^r)^2}.$$

If m is even, then the two sums cancel out and we end up with $\frac{1}{2} g_0(\tau)$ for the constant term which is certainly in $\mathcal{QM}$ by Proposition 1.1. If m is odd, then we get a contribution from the sum

$$\sum_{r\geq 1} \frac{r^s q^r}{(1-q^r)^2},$$

where $s := m - 2\ell - 1 \geq 0$ is even. For $s = 0$, this sum equals $\frac{1}{24}(1 - E_2(\tau)) \in \mathcal{QM}$. For $s \geq 2$, the sum is the first $\left(q\frac{\mathrm{d}}{\mathrm{d}q}\right)$-derivative of the Eisenstein series $E_{s-1}(\tau)$ and thus in $\mathcal{QM}$.

For the induction step, we need [41, formula (11)] that allows us to express J_1^ℓ as a linear combination of J_ℓ, an elliptic Jacobi form K_ℓ, and quasi-Jacobi forms involving J_1^k, $k < \ell$, for which the induction hypothesis applies. Thus in order to finish the proof we have to argue that

$$\mathrm{CT}_\zeta J_\ell(z;\tau)g(z;\tau) \in \mathcal{QM},$$

where $g(z;\tau)$ is elliptic. As in the $j = 1$ case, we first observe that

$$\frac{1}{\ell}(B_\ell - J_\ell(z;\tau)) = \sum_{n\geq 1}\left(\frac{q^n p_\ell(q^n)\zeta^n}{(1-q^n)^\ell} + (-1)^\ell \frac{q^n p_\ell(q^n)\zeta^{-n}}{(1-q^n)^\ell}\right),$$

where p_ℓ are as above. We distinguish two cases: ℓ is even and ℓ is odd. Assume first that $\ell = 2k$ is even. After we multiply out Fourier expansions and extract the constant term as above, the constant term takes the shape

$$\sum_{r\geq 1}\frac{q^{2r}r^s p_{2k}(q^r)}{(1-q^r)^{2k+1}} - (-1)^s \sum_{r\geq 1}\frac{q^r r^s p_{2k}(q^r)}{(1-q^r)^{2k+1}},$$

where $s := m - 2n - 1$ could be either even or odd. For s even, we have

$$\sum_{r\geq 1}\frac{r^s q^{2r} p_{2k}(q^r)}{(1-q^r)^{2k+1}} - (-1)^s \sum_{r\geq 1}\frac{r^s q^r p_{2k}(q^r)}{(1-q^r)^{2k+1}} = -\sum_{r\geq 1}\frac{r^s q^r p_{2k}(q^r)}{(1-q^r)^{2k}}$$

whereas for s odd, we obtain

$$\sum_{r\geq 1}\frac{r^s q^{2r} p_{2k}(q^r)}{(1-q^r)^{2k+1}} + \sum_{r\geq 1}\frac{r^s q^r p_{2k}(q^r)}{(1-q^r)^{2k+1}} = \sum_{r\geq 1}\frac{r^s q^r (1+q^r) p_{2k}(q^r)}{(1-q^r)^{2k+1}}.$$

By Lemma 1.2, both series are in $\mathcal{QM}$. Similarly we argue for ℓ odd. That completes the proof. □

Denote by $D_z = \frac{1}{2\pi \mathrm{i}}\frac{\partial}{\partial z}$ and $D_\tau = \frac{1}{2\pi \mathrm{i}}\frac{\partial}{\partial \tau}$, and let $s(x,y) \in \mathbb{C}[x,y]$ be a polynomial with complex coefficients.

Corollary 1.4. *Let $f(z;\tau) \in \mathcal{QJ}^0$. Then*

$$\mathrm{CT}_\zeta(s(D_z, D_\tau)f(z;\tau)) \in \mathcal{QM}.$$

Proof. We only have to argue that the ring $\mathcal{QJ}^0$ is closed under D_z and D_τ. This can be verified explicitly on the generators J_1, $\wp$, $\wp'$, E_4, E_6 (see, for instance, [42, 50]). □

An important special case involves products of derivatives of the P-functions introduced earlier. A similar result appeared previously in [26, Proposition 5.9], though our proof differs.

Proposition 1.5. *We have*

(*i*) *For $m \geq 1$ and $k \geq 1$,*

$$\mathrm{CT}_\zeta\big(P_k(\zeta;\tau)^m\big) \in \mathcal{QM}.$$

(*ii*) *More generally, for every polynomial $s(x,y)$ and $k \geq 1$,*

$$\mathrm{CT}_\zeta\big\{(s(D_\tau, D_z)P_k(\zeta;\tau))^m\big\} \in \mathcal{QM}.$$

Proof. Part (ii) follows immediately from Theorem 1.3, Corollary 1.4, and formulas (1.2)–(1.4). □

1.3. **Characters of genus zero class-$\mathcal{S}$ vertex operator algebras.** In this section, $\mathfrak{g}$ denotes a finite dimensional Lie algebra of ADE type for simplicity. In the introduction, we constructed a q-MZV function associated with $\mathfrak{g}$, $\zeta_{\mathfrak{g},q}(k_\alpha;\alpha\in\Delta_+)$. Letting $k_\alpha\geq 1$, we consider a slightly different q-MZV model where all k_α are equal and there is an extra polynomial dependence on the labels of $\lambda\in P_+$:

$$\zeta^s_{\mathfrak{g},q}(k):=\sum_{\lambda\in P_+}(\dim(V(\lambda))^s\frac{q^{k\langle\lambda+\rho,\rho\rangle}}{\prod_{\alpha\in\Delta_+}(1-q^{\langle\lambda,\alpha+\rho\rangle})^k}.$$

The boldface symbol $\mathbf{V}_{\mathfrak{g},k}$ denotes Arakawa's vertex operator algebra attached to the Lie algebra $\mathfrak{g}$ (actually our notation $\mathbf{V}_{\mathfrak{g},k}$ corresponds to Arakawa's $\mathbf{V}^{\mathcal{S}}_{G,b}$ with $\mathfrak{g}=\mathrm{Lie}(G)$ and $b=k+2$; see [5]) and the positive integer parameter k; it should not be confused with the ordinary italic letter k used as a summation or weight parameter elsewhere. Here $V(\lambda)$ is the finite-dimensional $\mathfrak{g}$-module of highest weight λ, so that $\dim(V(\lambda))$ can be computed using the Weyl dimension formula and $\rho=\frac{1}{2}\sum_{\alpha\in\Delta_+}\alpha$ is the half-sum of positive roots. Then we have a remarkable result of Arakawa on class-$\mathcal{S}$ vertex operator algebras motivated by the previous very important work of Beem, Rastelli and others [10, 11].

Theorem 1.6 ([5])**.** *For every $k\geq 3$, there is a vertex operator algebra $\mathbf{V}_{\mathfrak{g},k}$ such that*

$$\mathrm{ch}[\mathbf{V}_{\mathfrak{g},k}](\tau)=\frac{1}{\eta(\tau)^{m_{\mathfrak{g},k}}}\zeta^k_{\mathfrak{g},q}(k-2), \tag{1.7}$$

where $m_{\mathfrak{g},k}=\dim(\mathfrak{g})k-(k-2)\mathrm{rank}(\mathfrak{g})$ is a positive integer and $\eta(\tau)=q^{1/24}\prod_{i\geq 1}(1-q^i)$ is the Dedekind η function.

Remark 1.7. Although this seems to be implicit in the literature, it is expected that

$$\dim X_{\mathbf{V}_{\mathfrak{g},k}}=m_{\mathfrak{g},k},$$

where $X_{\mathbf{V}_{\mathfrak{g},k}}$ is the associated variety of $\mathbf{V}_{\mathfrak{g},k}$. In the 4d/2d correspondence, the associated variety is expected to recover the Higgs branch of the corresponding four-dimensional theory, while the vertex operator algebra character is identified with its Schur index. Thus formula (1.7) suggests that the generalized q-MZV is the infinite-dimensional analogue of the numerator in a Hilbert-series presentation: it contains the non-free contribution after the universal product factor has been separated. We use the phrase "Hilbert-series numerator" here, rather than "h-series", to avoid confusion with the h-polynomial of a finitely generated graded algebra.

Example 1.8 ($\mathfrak{g}=\mathfrak{sl}_2$)**.** For $k\geq 3$,

$$\zeta^k_{\mathfrak{sl}_2,q}(k-2)=\sum_{n\geq 1}\frac{n^k q^{\frac{n}{2}(k-2)}}{(1-q^n)^{k-2}}.$$

Example 1.9 ($\mathfrak{g}=\mathfrak{sl}_3$)**.** For $k\geq 3$,

$$\zeta^k_{\mathfrak{sl}_3,q}(k-2)=\sum_{m_1,m_2\geq 1}(m_1m_2(m_1+m_2))^k\,\frac{q^{(m_1+m_2)(k-2)}}{(1-q^{m_1})^{k-2}(1-q^{m_2})^{k-2}(1-q^{m_1+m_2})^{k-2}}.$$

It is not hard to see that we also have

$$\zeta^k_{\mathfrak{sl}_{n+1},q}(k-2)=\sum_{m_1,\ldots,m_n\geq 1}\left(\prod_{1\leq i\leq j\leq n}(m_i+\cdots+m_j)^k\right)\cdot\frac{q^{(k-2)(\frac{1}{2}\sum_{i=1}^n(i-1+1)(n-i+1)m_i)}}{\prod_{1\leq i\leq j\leq n}(1-q^{m_i+\cdots+m_j})^{k-2}}.$$

This can be rewritten as

$$\zeta^k_{\mathfrak{sl}_{n+1},q}(k-2)=\sum_{m_1,\dots,m_n\geq 1}\prod_{1\leq i\leq j\leq n}\left((m_i+\cdots+m_j)^k\frac{q^{\frac{1}{2}(k-2)(im_i+\cdots+jm_j)}}{(1-q^{m_i+\cdots+m_j})^{k-2}}\right).$$

Based on the $\mathfrak{sl}_2$ case studied by Beem and Rastelli [11], it is expected the following.

Conjecture 1.10. *We have*

(1) *For $k\geq 3$ even, $\zeta^k_{\mathfrak{sl}_{n+1},q}(k-2)$ belongs to $\mathcal{QM}$ and for $k\geq 3$ odd it belongs to $\mathcal{QM}[2]:=\mathbb{Q}\big[E_2(\tau),\theta_2(\tau)^4,\theta_3(\tau)^4\big]$.*[2]
(2) *For $k\geq 3$, the character $\mathrm{ch}[\mathbf{V}_{\mathfrak{g},k}](\tau)$ is a solution of a modular linear differential equation (MLDE) with coefficients in $\mathcal{QM}$ or $\mathcal{QM}[2]$, depending on the parity of k.*

Beem and Rastelli verified these conjectures for numerous examples for $\mathfrak{g}=\mathfrak{sl}_2$ (for both k even and odd) and also for additional examples coming from genus $g\geq 1$ theories. These higher-genus theories are not discussed in Arakawa's paper but presumably correspond to characters of (mostly unknown) vertex operator algebras due to 2d/4d correspondences [10]. In [16], we proved a broader theorem that implies the even-parameter quasi-modularity assertion in Conjecture 1.10 (1) for the relevant A-type series. It does not by itself prove the MLDE assertion in Conjecture 1.10 (2). Theorem 1.13 below supplies a direct, self-contained constant-term proof for the first non-trivial case $\mathfrak{g}=\mathfrak{sl}_3$, while Proposition 3.14 treats the corresponding symmetrized family.

1.4. **Bi-brackets of multiple q-zeta values of a Lie algebra $\mathfrak{g}$.** The notation in this subsection is independent of the one-variable zeta values in (0.1); here the square-bracket notation records two arrays of parameters, following the terminology of bi-brackets. Motivated by Arakawa's character formula, we can also define more general multiple q-zeta series, reminiscent of the *bi-brackets* construction in the setup of q-MZVs [7, 8, 9]. For simplicity, we again only discuss the A-type here so the positive roots are parametrized by the segments $[i,j]$, $i\leq j$.

Let $k_{[i,j]}\geq 1$, $1\leq i\leq j$ and $s_{[i,j]}\geq 0$ be integers, and

$$\begin{aligned}&\zeta_{\mathfrak{sl}_{n+1},q}\begin{bmatrix}k_{[i,j]};&1\leq i\leq j\leq n\\ s_{[i,j]};&1\leq i\leq j\leq n\end{bmatrix}\\&\qquad:=\sum_{m_1,\dots,m_n\geq 1}\prod_{1\leq i\leq j\leq n}(m_i+\cdots+m_j)^{s_{[i,j]}}\frac{q^{\frac{1}{2}k_{[i,j]}(im_i+\cdots+jm_j)}}{(1-q^{m_i+\cdots+m_j})^{k_{[i,j]}}}.\end{aligned}$$

With this definition, we have

$$\zeta_{\mathfrak{sl}_{n+1},q}\begin{bmatrix}k_{[i,j]}=k;&1\leq i\leq j\leq n\\ s_{[i,j]}=k+2;&1\leq i\leq j\leq n\end{bmatrix}=\zeta^{k+2}_{\mathfrak{sl}_{n+1},q}(k).$$

Based on many examples, we expect a symmetric formula to hold.

Conjecture 1.11. *Assume that all $k_{[i,j]}$ are even and $s=s_{[i,j]}$ for all $[i,j]$. Then*

$$\sum_{\sigma\in W}\zeta_{\mathfrak{sl}_{n+1},q}\left[k_{\sigma[1,1]},k_{\sigma[1,2]},\dots,k_{\sigma[n,n]};s\right]\in\mathcal{QM},$$

where $k_{-[i,j]}=k_{[i,j]}$.

[2]It is possible to formulate a slightly stronger conjecture depending on whether or not $\rho\in Q$, where Q is the root lattice of $\mathfrak{g}$.

One can also formulate a similar conjecture for other root systems. We also expect quasi-modularity to hold for $\zeta_{\mathfrak{sl}_{n+1},q}\left[\begin{smallmatrix} k_{[i,j]};\ 1\le i\le j\le n \\ s_{[i,j]};\ 1\le i\le j\le n \end{smallmatrix}\right]$ provided that k_α and s_α have the same parity.

1.5. **Example: Quasi-modularity for $\mathfrak{sl}(3)$.** The quasi-modularity of $\mathfrak{sl}(2)$ q-MZV values and their modifications is well known to experts, and therefore we only briefly comment on this case. Recall the Lie-theoretic q-multiple zeta values introduced in (0.1). In the simplest case $\mathfrak{g} = \mathfrak{sl}_2$, one has

$$\zeta_{\mathfrak{sl}_2,q}(2k) = \sum_{n\ge 1} \frac{q^{nk}}{(1-q^n)^{2k}}.$$

It is well known that, up to an additive constant, these Lambert series are linear combinations of Eisenstein series and consequently,

$$\zeta_{\mathfrak{sl}_2,q}(2k) \in \mathcal{QM}.$$

Similarly, for every $k \ge 1$, there exist constants $c_{k,i} \in \mathbb{Q}$, $1 \le i \le k$, such that, for every $n \ge 1$ and $a \ge 0$, the following identity of rational functions in q holds:

$$\frac{n^{2k+a}q^{nk}}{(1-q^n)^{2k}} = \sum_{i=1}^{k} c_{k,i} \left(q\frac{\mathrm{d}}{\mathrm{d}q}\right)^{2k+1-2i} \frac{n^{2i-1+a}q^n}{1-q^n}. \tag{1.8}$$

Summing over n and letting $a = 2$ immediately gives

$$\zeta^{2k+2}_{\mathfrak{sl}_2,q}(2k) = \sum_{i=1}^{k} c_{k,i} \left(q\frac{\mathrm{d}}{\mathrm{d}q}\right)^{2k+1-2i} E_{2+2i}(q) \in \mathbb{Q}[E_2, E_4, E_6].$$

Corollary 1.12. *For integers $k \ge 1$ and $a \ge 0$, we have*

$$\begin{aligned} &\sum_{i=1}^{k} c_{k,i} D_\tau^{2k+1-2i} \left(\sum_{n\ge 1} \left(\frac{n^{2i-1+a}q^n\zeta^n}{1-q^n} + (-1)^{a+1} \frac{n^{2i-1+a}q^n\zeta^{-n}}{1-q^n} \right) \right) \\ &\qquad = \sum_{n\ge 1} \left(\frac{n^{2k+a}q^{nk}\zeta^n}{(1-q^n)^{2k}} + (-1)^a \frac{n^{2k+a}q^{nk}\zeta^{-n}}{(1-q^n)^{2k}} \right). \end{aligned} \tag{1.9}$$

Equipped with (1.9) and results from Section 1.3, we are now ready to prove the following.

Theorem 1.13. *Let $k \ge 1$ and let $a \ge 0$ be even. Then*

$$\sum_{m_1,m_2\ge 1} (m_1m_2(m_1+m_2))^{2k+a} \frac{q^{2k(m_1+m_2)}}{(1-q^{m_1})^{2k}(1-q^{m_2})^{2k}(1-q^{m_1+m_2})^{2k}} \in \mathcal{QM}.$$

In particular, taking $a = 2$ gives $\zeta^{2k+2}_{\mathfrak{sl}_3,q}(2k) \in \mathcal{QM}$.

Proof. Keeping a even, we prove that

$$\zeta^{2k+a}_{\mathfrak{sl}_3,q}(2k) = \sum_{m_1,m_2\ge 1} (m_1m_2(m_1+m_2))^{2k+a} \frac{q^{2k(m_1+m_2)}}{(1-q^{m_1})^{2k}(1-q^{m_2})^{2k}(1-q^{m_1+m_2})^{2k}}$$

also belongs to $\mathcal{QM}$. Keeping a even, observe an identity

$$\begin{aligned}
&\mathrm{CT}_\zeta\left(\sum_{n\geq 1}\frac{n^{2k+a}q^{nk}\zeta^n}{(1-q^n)^{2k}}+(-1)^a\frac{n^{2k+a}q^{nk}\zeta^{-n}}{(1-q^n)^{2k}}\right)^3\\
&\quad=\mathrm{CT}_\zeta\left(\sum_{n\geq 1}\frac{n^{2k+a}q^{nk}\zeta^n}{(1-q^n)^{2k}}+\frac{n^{2k+a}q^{nk}\zeta^{-n}}{(1-q^n)^{2k}}\right)^3\\
&\quad=6\sum_{m_1,m_2\geq 1}(m_1m_2(m_1+m_2))^{2k+a}\,\frac{q^{2k(m_1+m_2)}}{(1-q^{m_1})^{2k}(1-q^{m_2})^{2k}(1-q^{m_1+m_2})^{2k}},
\end{aligned}$$

where in the last step we expanded the triple product as triple sums and we extract the constant term from each individual term. Formula (1.9), together with Proposition 1.5, now gives the assertion. □

References

[1] Adamović D., Milas A., On some vertex algebras related to $V_{-1}(\mathfrak{sl}(n))$ and their characters, *Transform. Groups* **26** (2021), 1–30, arXiv:1805.09771.

[2] Andrews G.E., The theory of partitions, *Encyclopedia Math. Appl.*, Vol. 2, Addison-Wesley Publishing Co., Reading, Mass., 1976.

[3] Andrews G.E., Stacked lattice boxes, *Ann. Comb.* **3** (1999), 115–130.

[4] Andrews G.E., Rose S.C.F., MacMahon's sum-of-divisors functions, Chebyshev polynomials, and quasi-modular forms, *J. Reine Angew. Math.* **676** (2013), 97–103, arXiv:1010.5769.

[5] Arakawa T., Chiral algebras of class $\mathcal{S}$ and Moore–Tachikawa symplectic varieties, arXiv:1811.01577.

[6] Arakawa T., Kawasetsu K., Quasi-lisse vertex algebras and modular linear differential equations, in Lie Groups, Geometry, and Representation Theory, *Progr. Math.*, Vol. 326, Birkhäuser, Cham, 2018, 41–57, arXiv:1610.05865.

[7] Bachmann H., Kühn U., A short note on a conjecture of Okounkov about a q-analogue of multiple zeta values, arXiv:1407.6796.

[8] Bachmann H., Kühn U., The algebra of generating functions for multiple divisor sums and applications to multiple zeta values, *Ramanujan J.* **40** (2016), 605–648, arXiv:1309.3920.

[9] Bachmann H., Kühn U., A dimension conjecture for q-analogues of multiple zeta values, in Periods in Quantum Field Theory and Arithmetic, *Springer Proc. Math. Stat.*, Vol. 314, Springer, Cham, 2020, 237–258, arXiv:1708.07464.

[10] Beem C., Lemos M., Liendo P., Peelaers W., Rastelli L., van Rees B.C., Infinite chiral symmetry in four dimensions, *Comm. Math. Phys.* **336** (2015), 1359–1433, arXiv:1312.5344.

[11] Beem C., Rastelli L., Vertex operator algebras, Higgs branches, and modular differential equations, *J. High Energy Phys.* **2018** (2018), no. 8, 114, 72 pages, arXiv:1707.07679.

[12] Beem C., Singh P., Razamat S.S., Schur indices of class $\mathcal{S}$ and quasimodular forms, *Phys. Rev. D* **105** (2022), 085009, 15 pages, arXiv:2112.10715.

[13] Bradley D.M., Multiple q-zeta values, *J. Algebra* **283** (2005), 752–798, arXiv:math/0402093.

[14] Bringmann K., Folsom A., Mahlburg K., Quasimodular forms and $s\ell(m|m)^\wedge$ characters, *Ramanujan J.* **36** (2015), 103–116, arXiv:1302.4040.

[15] Bringmann K., Jennings-Shaffer C., Milas A., Graph schemes, graph series, and modularity, *J. Combin. Theory Ser. A* **197** (2023), 105749, 37 pages, arXiv:2105.05660.

[16] Bringmann K., Milas A., van Ittersum J.W.M., Quasimodularity of q-series associated to root lattices, in preparation.

[17] Calinescu C., Lepowsky J., Milas A., Vertex-algebraic structure of the principal subspaces of level one modules for the untwisted affine Lie algebras of types A, D, E, *J. Algebra* **323** (2010), 167–192, arXiv:0908.4054.

[18] Carlsson E., Vertex operators, Grassmannians, and Hilbert schemes, *Comm. Math. Phys.* **300** (2010), 599–613, arXiv:0910.5528.

[19] Dong C., Mason G., Nagatomo K., Quasi-modular forms and trace functions associated to free boson and lattice vertex operator algebras, *Int. Math. Res. Not.* **2001** (2001), 409–427, arXiv:math/0004132.

[20] Dotsenko V., Feigin E., Reineke M., Koszul algebras and Donaldson–Thomas invariants, *Lett. Math. Phys.* **112** (2022), 106, 39 pages, arXiv:2111.07588.

[21] Dotsenko V., Mozgovoy S., DT invariants from vertex algebras, *J. Inst. Math. Jussieu* **24** (2025), 291–339, arXiv:2108.10338.
[22] Efimov A.I., Cohomological Hall algebra of a symmetric quiver, *Compos. Math.* **148** (2012), 1133–1146, arXiv:1103.2736.
[23] Ekholm T., Gruen A., Gukov S., Kucharski P., Park S., Stošić M., Sułkowski P., Branches, quivers, and ideals for knot complements, *J. Geom. Phys.* **177** (2022), 104520, 75 pages, arXiv:2110.13768.
[24] Feigin B., Stoyanovsky A.V., Quasi-particles models for the representations of Lie algebras and geometry of flag manifold, arXiv:hep-th/9308079.
[25] Goncharov A.B., Multiple polylogarithms, cyclotomy and modular complexes, *Math. Res. Lett.* **5** (1998), 497–516.
[26] Goujard E., Möller M., Counting Feynman-like graphs: Quasimodularity and Siegel–Veech weight, *J. Eur. Math. Soc. (JEMS)* **22** (2020), 365–412, arXiv:1609.01658.
[27] Herzog J., Hibi T., Distributive lattices, bipartite graphs and Alexander duality, *J. Algebraic Combin.* **22** (2005), 289–302, arXiv:math/0307235.
[28] Hoffman M.E., Algebraic aspects of multiple zeta values, in Zeta Functions, Topology and Quantum Physics, *Dev. Math.*, Vol. 14, Springer, New York, 2005, 51–73, arXiv:math/0309425.
[29] Ihara K., Kaneko M., Zagier D., Derivation and double shuffle relations for multiple zeta values, *Compos. Math.* **142** (2006), 307–338.
[30] Jennings-Shaffer C., Milas A., On q-series identities for false theta series, *Adv. Math.* **375** (2020), 107411, 22 pages, arXiv:2001.11368.
[31] Jennings-Shaffer C., Milas A., Further q-series identities and conjectures relating false theta functions and characters, in Lie groups, Number Theory, and Vertex Algebras, *Contemp. Math.*, Vol. 768, American Mathematical Society, Providence, RI, 2021, 253–269, arXiv:2005.13620.
[32] Kawasetsu K., The free generalized vertex algebras and generalized principal subspaces, *J. Algebra* **444** (2015), 20–51, arXiv:1502.05276.
[33] Kucharski P., Reineke M., Stošić M., Sułkowski P., Knots-quivers correspondence, *Adv. Theor. Math. Phys.* **23** (2019), 1849–1902, arXiv:1707.04017.
[34] Lang S., Elliptic functions, 2nd ed., *Grad. Texts in Math.*, Vol. 112, Springer, New York, 1987.
[35] Li H., Some remarks on associated varieties of vertex operator superalgebras, *Eur. J. Math.* **7** (2021), 1689–1728, arXiv:2007.04522.
[36] Li H., Milas A., Jet schemes, quantum dilogarithm and Feigin–Stoyanovsky's principal subspaces, *J. Algebra* **640** (2024), 21–58, arXiv:2010.02143.
[37] Libgober A., Elliptic genera, real algebraic varieties and quasi-Jacobi forms, in Topology of Stratified Spaces, *Math. Sci. Res. Inst. Publ.*, Vol. 58, Cambridge University Press, Cambridge, 2011, 95–120, arXiv:0904.1026.
[38] Milas A., Formal differential operators, vertex operator algebras and zeta-values, II, *J. Pure Appl. Algebra* **183** (2003), 191–244, arXiv:math/0303154.
[39] Milas A., On certain automorphic forms associated to rational vertex operator algebras, in Moonshine: The First Quarter Century and Beyond, *London Math. Soc. Lecture Note Ser.*, Vol. 372, Cambridge University Press, Cambridge, 2010, 330–357.
[40] Milas A., Penn M., Lattice vertex algebras and combinatorial bases: general case and $\mathcal{W}$-algebras, *New York J. Math.* **18** (2012), 621–650.
[41] Oberdieck G., A Serre derivative for even weight Jacobi forms, arXiv:1209.5628.
[42] Oberdieck G., Gromov–Witten invariants of the Hilbert schemes of points of a K3 surface, *Geom. Topol.* **22** (2018), 323–437, arXiv:1406.1139.
[43] Ohno Y., Okuda J., Zudilin W., Cyclic q-MZSV sum, *J. Number Theory* **132** (2012), 144–155.
[44] Ohno Y., Zagier D., Multiple zeta values of fixed weight, depth, and height, *Indag. Math. (N.S.)* **12** (2001), 483–487.
[45] Okunkov A., Hilbert schemes and multiple q-zeta values, *Funct. Anal. Appl.* **48** (2014), 138–144, arXiv:1404.3873.
[46] The on-line encyclopedia of integer sequences, `https://oeis.org/`.
[47] Pan Y., Peelaers W., Exact Schur index in closed form, *Phys. Rev. D* **106** (2022), 045017, 34 pages, arXiv:2112.09705.
[48] Qin Z., Yu F., On Okounkov's conjecture connecting Hilbert schemes of points and multiple q-zeta values, *Int. Math. Res. Not.* **2018** (2018), 321–361, arXiv:1510.00837.
[49] Renteln P., The Hilbert series of the face ring of a flag complex, *Graphs Combin.* **18** (2002), 605–619.

[50] van Ittersum J.W.M., Partitions and quasimodular forms: Variations on the Bloch–Okounkov theorem, Ph.D. thesis, Utrecht University, 2021, https://staff.fnwi.uva.nl/j.w.m.vanittersum/files/theses/thesis_vanIttersum_withcover.pdf.
[51] Zagier D., Multiple zeta values, Preprint, 1995.
[52] Zhu Y., Modular invariance of characters of vertex operator algebras, *J. Amer. Math. Soc.* **9** (1996), 237–302.
[53] Zudilin W., Algebraic relations for multiple zeta values, *Russian Math. Surveys* **58** (2003), 1–29.
[54] Zudilin W., Multiple q-zeta brackets, *Mathematics* **3** (2015), 119–130, arXiv:1412.0163.